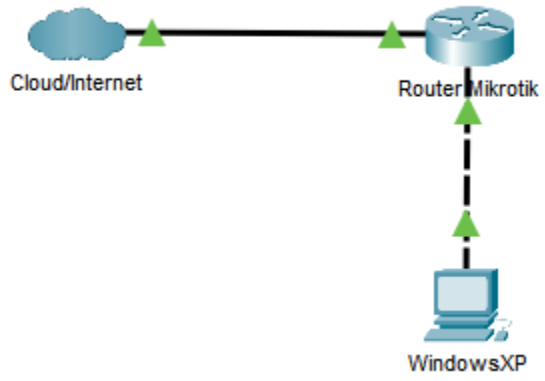
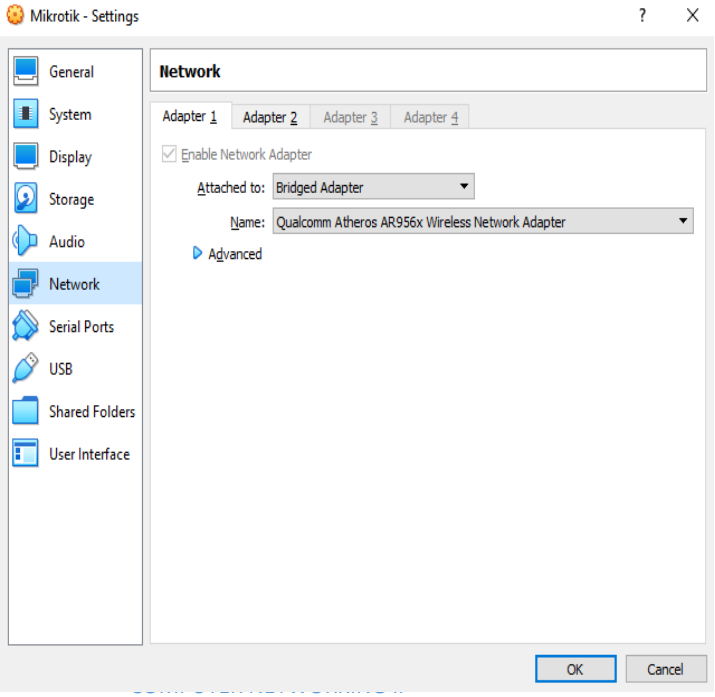
Konfigurasaun Baziku Iha Mikrotik Utiliza Virtualbox

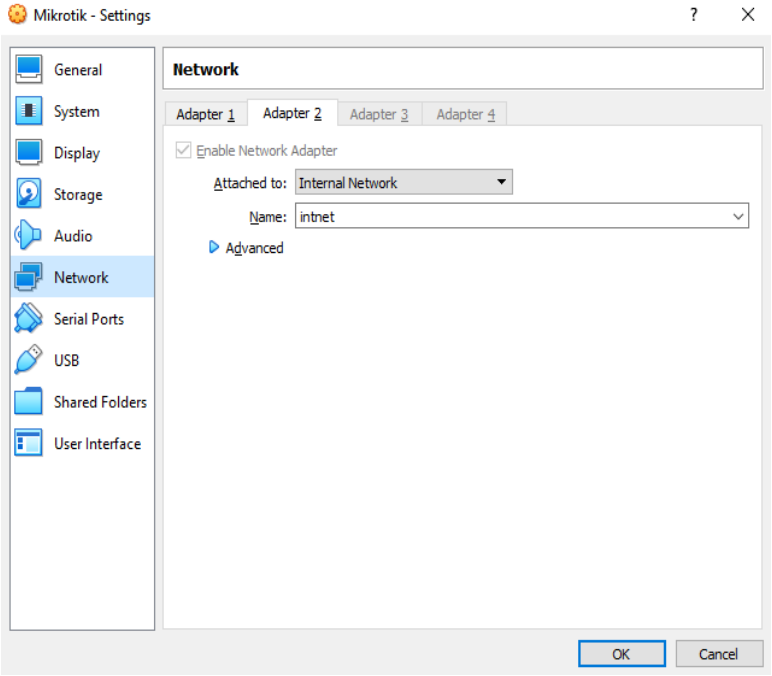
(IP Address, DNS, Route, NAT, DHCP no HOTSPOT)



Konfigurasaun VirtualBox

1. **Mikrotik**
   a. Klik iha **Pengaturan/Settings**, Konfigura parte **Network/Jaringan** uja adapter rua
      - Adapter 1 iha parte **attached to** hili **Bridge Adapter (Wireless Adapter)**
      - Adapter 2 iha parte **attached to** hili **Internal Netwrok (inet)**

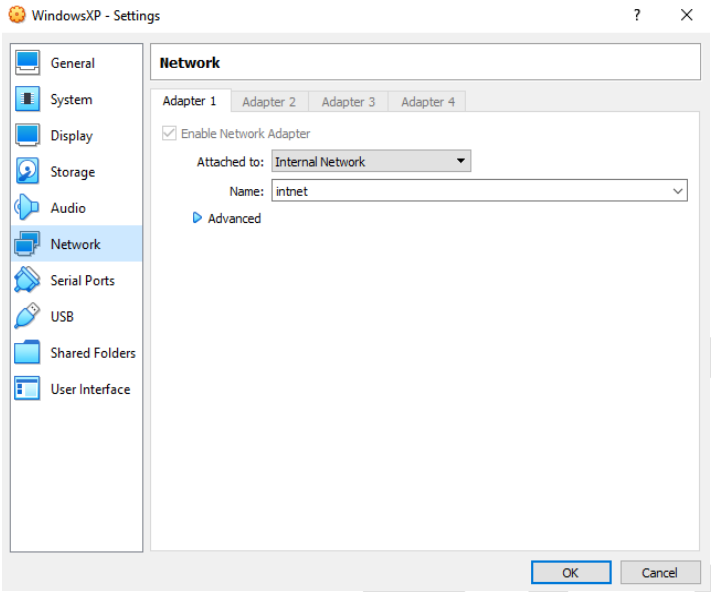




2. **Windows XP**

   a. Klik iha **Pengaturan/Settings**, Konfigura parte **Network/Jaringan** uja adapter ida

      - Adapter 1 iha parte **attached to** hili **Internal Network (inet)**



Konfigurasaun IP Address

1. **Mikrotik**

   a. Hakerek **Command line** tuir mai

      - ip address add address=192.168.1.1/24 interface=ether2
      - ip dhcp-client add interface=ether1 add-default-route=yes
      - ip dhcp-client enable number=0
      - ip address print

```
[admin@MikroTik] > ip address add address=192.168.100.1/24 interface=ether2
[admin@MikroTik] > ip dhcp-client add interface=ether1 add-default-route=yes
[admin@MikroTik] > ip dhcp-client enable numbers=0
[admin@MikroTik] > ip address print
Flags: X - disabled, I - invalid, D - dynamic
 #    ADDRESS            NETWORK         INTERFACE
 0    192.168.100.1/24   192.168.100.0   ether2
 1 D  192.168.252.100/24 192.168.252.0   ether1
[admin@MikroTik] >
```

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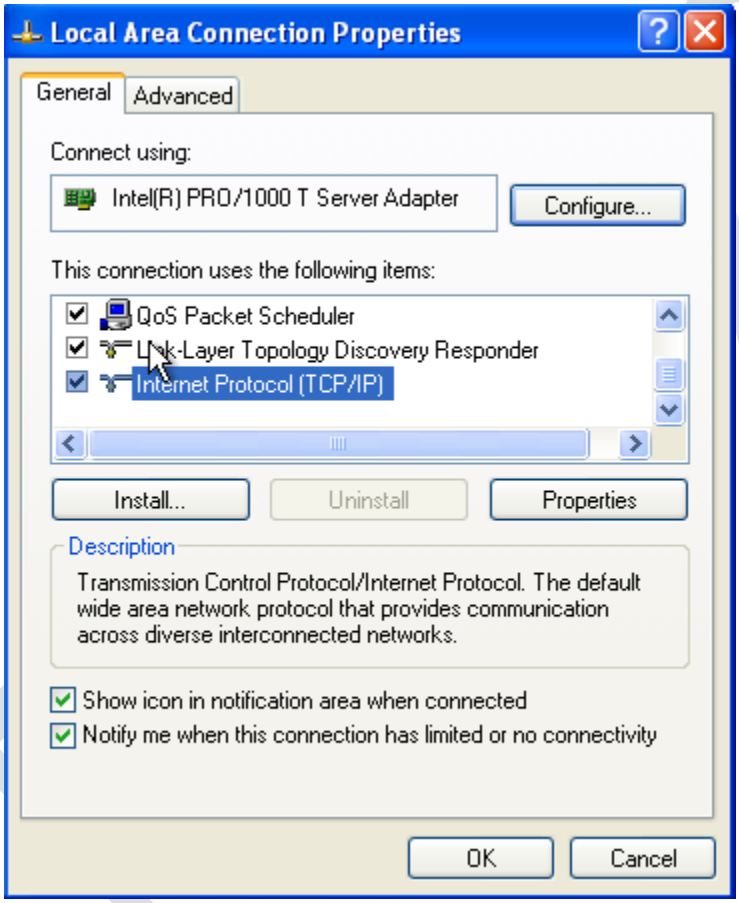
2. **Windows XP**
   a. Klik **Start → Control Panel**
   b. Hili **Network and Internet Connections**
   c. Pois klik **Network Connections**
   d. Double Click ou Klik dala rua iha icon **Local Area Connection**
   e. Klik **Properties**, sei mosu informasaun husi **Local Area Connection**
   f. Scroll tun hili **Internet Protocol(TCP/IP)**



   g. Klik **Properties**, sei mosu tan janela foun hodi prense IP Address
   h. Klik iha **Use the following Ip address**

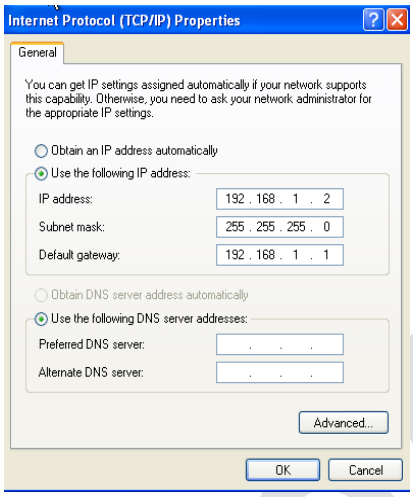
i. Prense ip address tuir mai

Ip address : 192.168.1.2

Subnetmask : 255.255.255.0

Default Gateway : 192.168.1.1

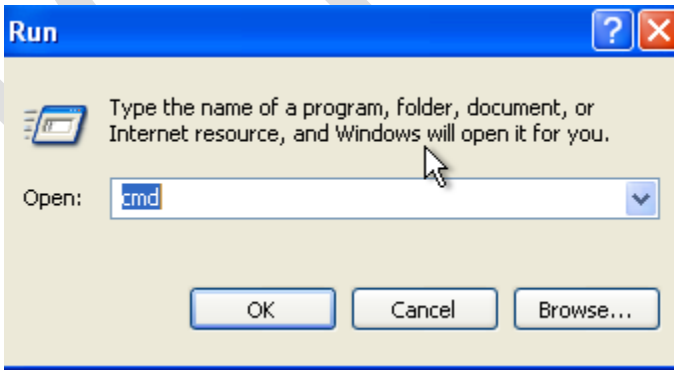
Pois klik **OK**



Koko Koneksaun

a. Iha **Windows Xp**

- Klik **Start** pois klik **RUN**
- Hakerek **cmd** pois Enter



- Hakerek comman line **ping 192.168.1.1** pois **enter**
- Kuandu mosu **replay** signifika katak Windows Xp konekta ona ho Mikrotik

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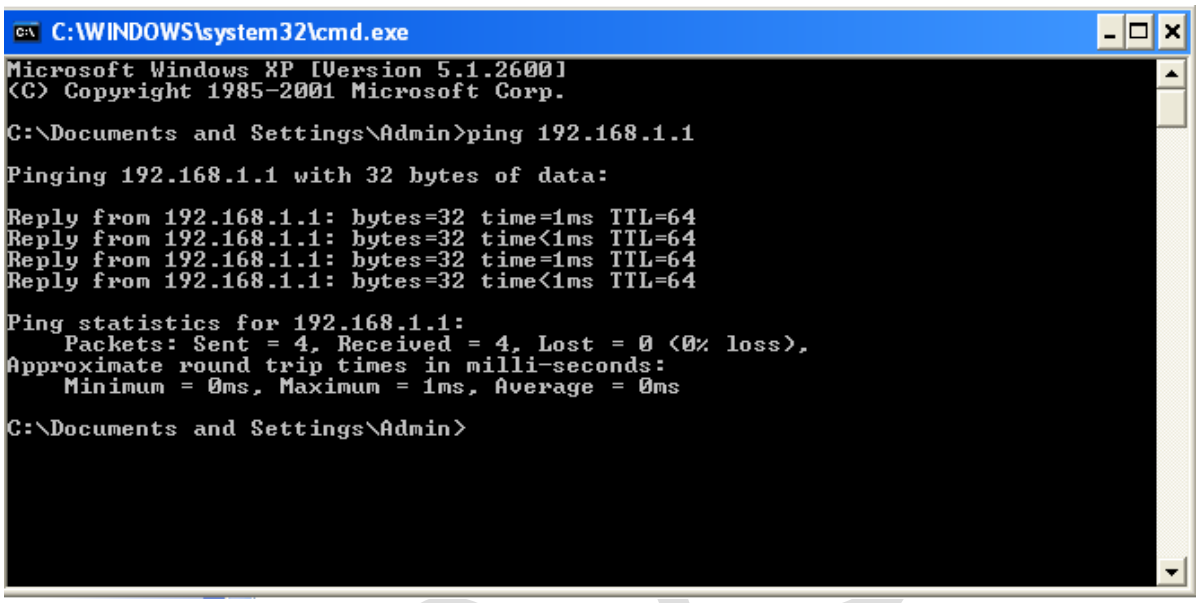
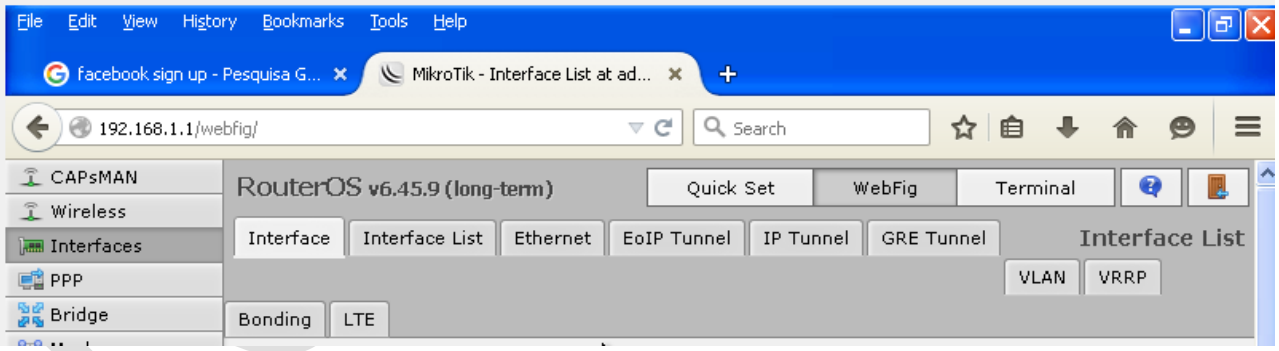


Konfigurasaun Mikrotik

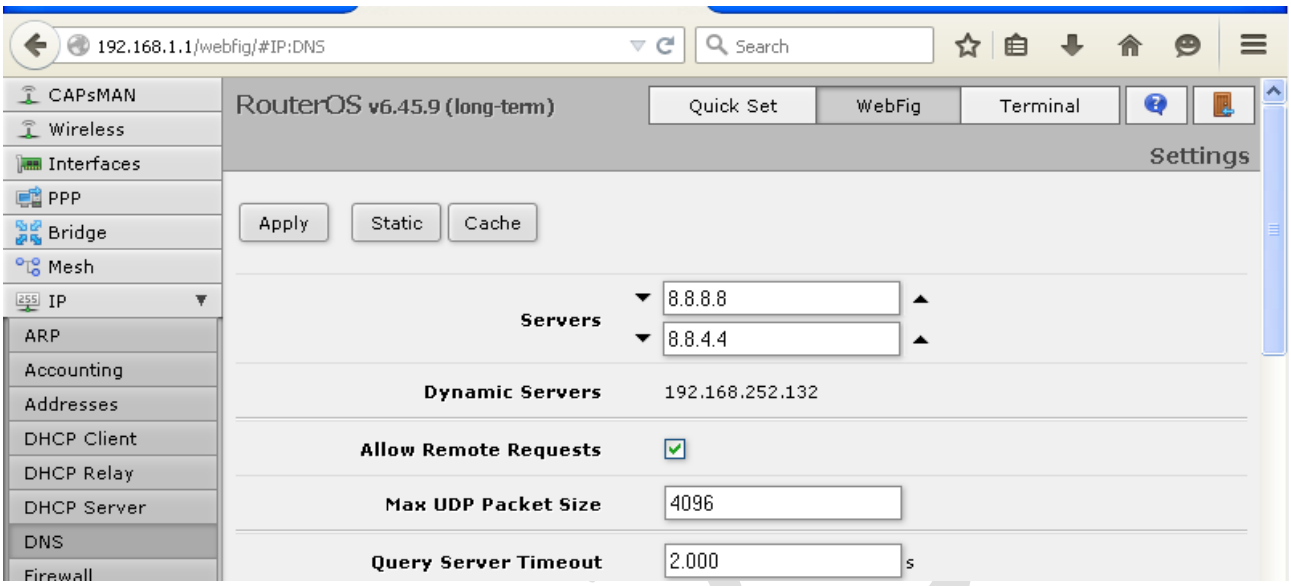
A. Asesu Webfig husi Mikrotik**,** bele mos uja aplikasi Winbox

1. Hakerek IP Address mikrotik iha browser (Mozilla, Chrome ou IE) pois **Enter**
2. Iha ne'e uja IP Address **192.168.1.1**



B. Konfigurasaun DNS

1. Iha **Mikrotik** klik menu **IP → DNS**
2. Aumenta Server husi DNS ho maneira hanehan anak-panah metan iha sorin los pois hakerek **8.8.8.8** no **8.8.4.4**
3. Fo marka iha **Allow Remote Requestes**
4. Klik **Apply** no **OK**

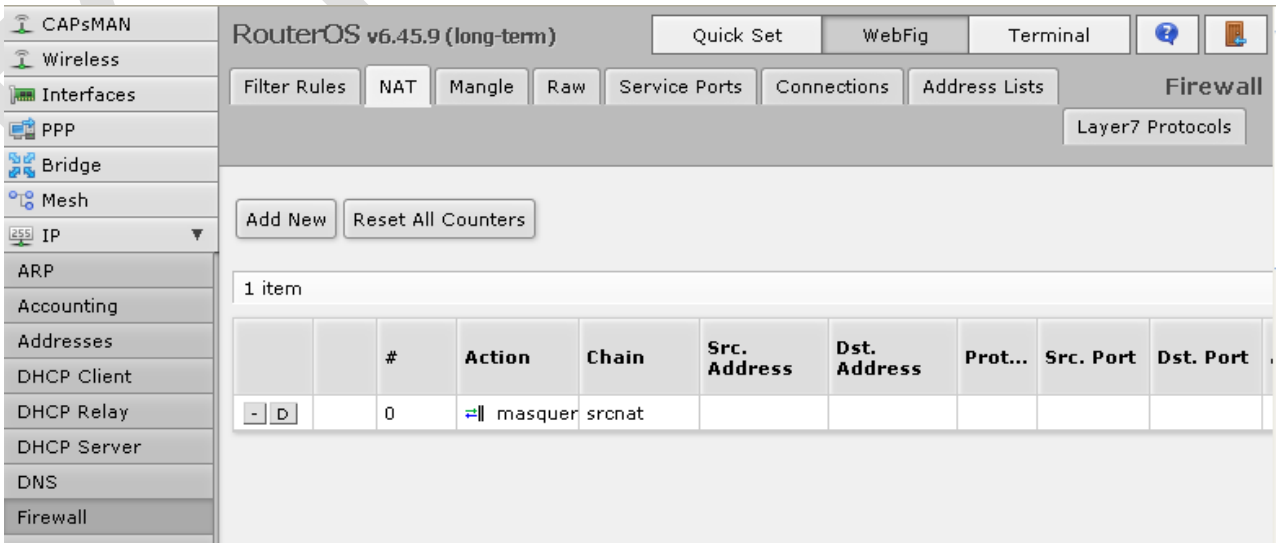


C. Konfigurasaun Routes

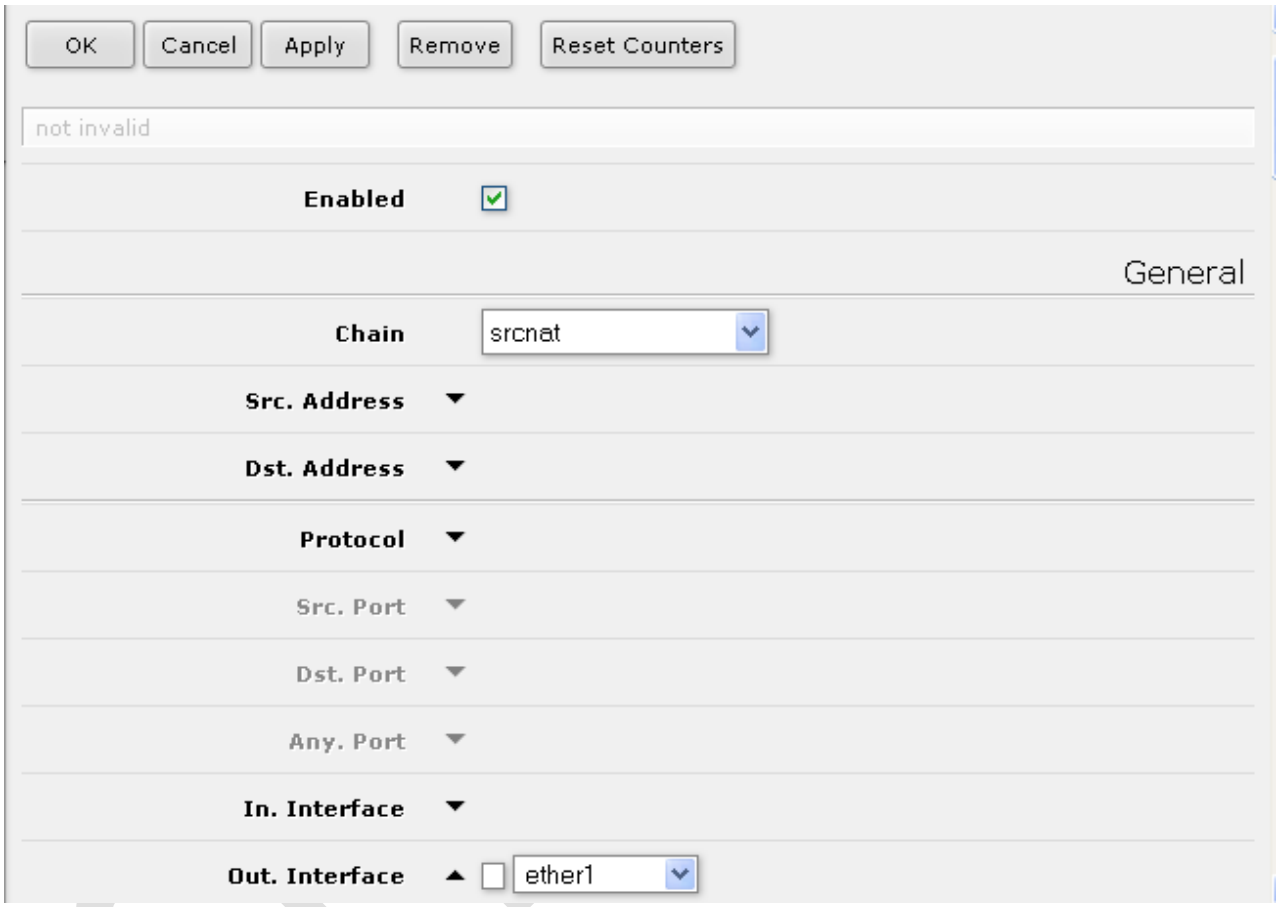
Karik internet uja Static IP address entaun presija konfigurasaun static route, tamba iha ne uja Dynamic IP Address husi ISP taun iha parte ida ne'e ita lapresija konfigura.

D. Konfigurasaun Firewall/NAT

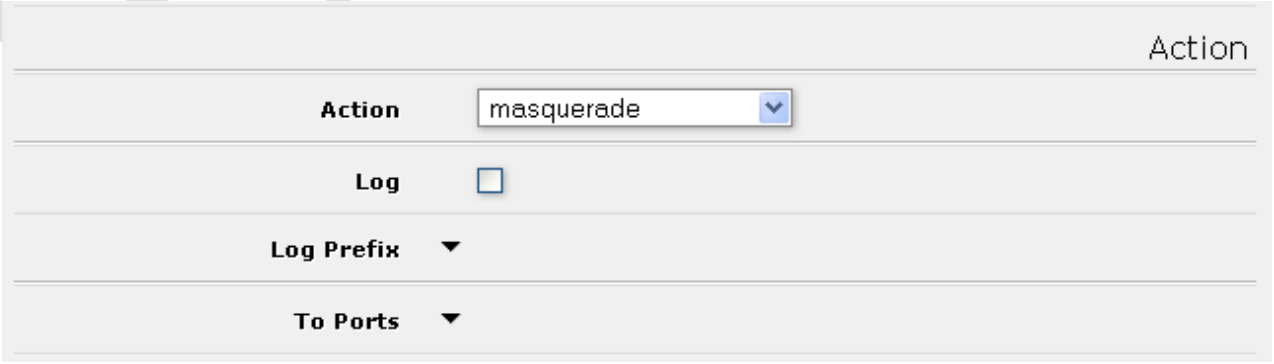
1. Iha **Mikrotik** klik menu **IP → Firewall**
2. Iha parte **Firewall** klik tab menu **NAT**
3. Pois klik **Add New**



4. Iha tab menu **General** parte **Chain** hili **srcnat**
5. Nafatin iha tab menu **General** iha parte **out interface** hili **ether1**



6. Ba fali iha tab menu **Action** iha parte **Action** hili **Masquerade**



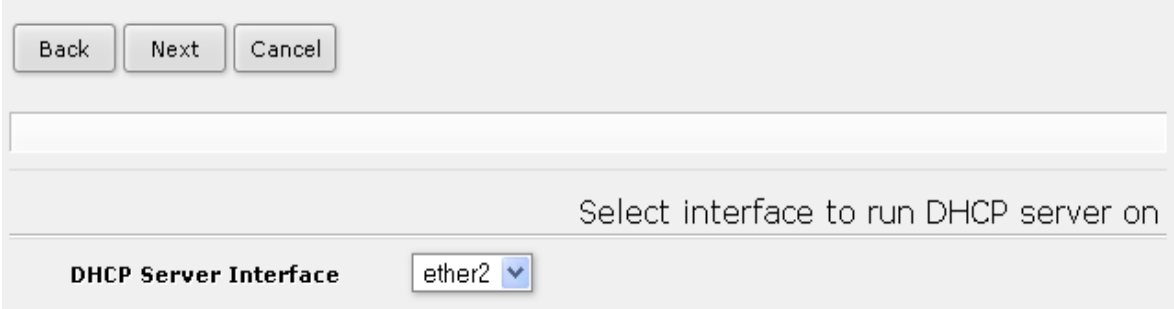
7. Klik **Apply** no **Ok**

E. Konfigurasaun DHCP Server

1. Iha **Mikrotik** klik menu **IP → DHCP Server**
2. Klik iha tab menu **DHCP Setup**
3. Hare tuir konfigurasaun tuir mai to remata:

   a. DHCP Server interface : ether2, klik **Next**

   (interface ne'ebe mak ita atu konfigura DHCP ba)

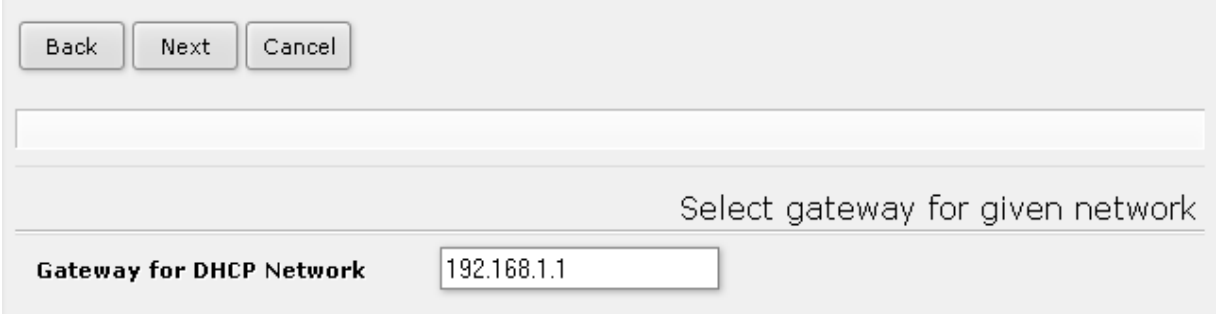


   b. DHCP Addres Space : 192.168.101/24, klik **Next**

   (IP Address ne'e mosu tuir IP Address ne'ebe antes ne'e konfigura ona)

   c. Gateway for DHCP Network : 192.168.1.1, klik **Next**

   (IP Address mikrotik ne'ebe mak ita konfigura ona)

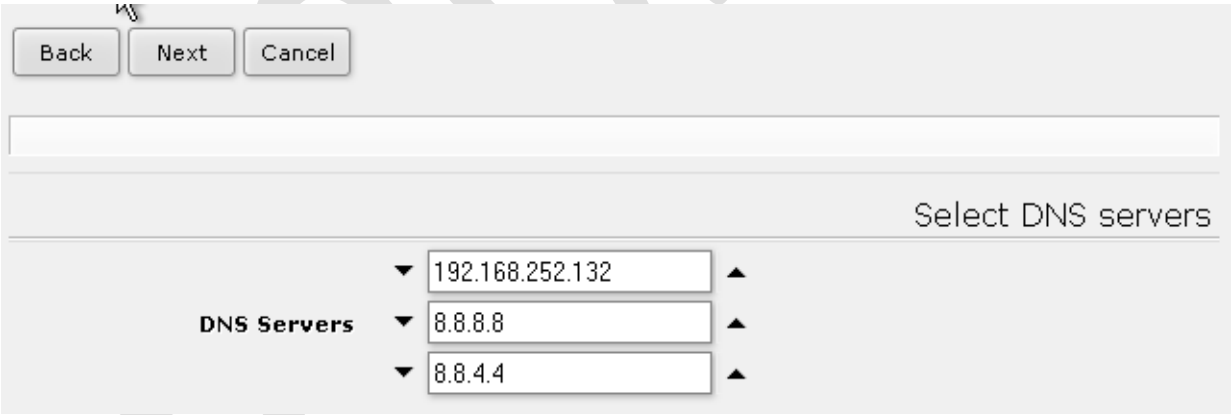


d. **Address to Give Out : 192.168.1.2 – 192.168.1.254,** klik **Next**

(Range IP Address sei fo tuir IP Address ne'ebe mak ita konfigura ona, tamba iha ne'e ita uja /24 maka Range husi /24 mak 254)

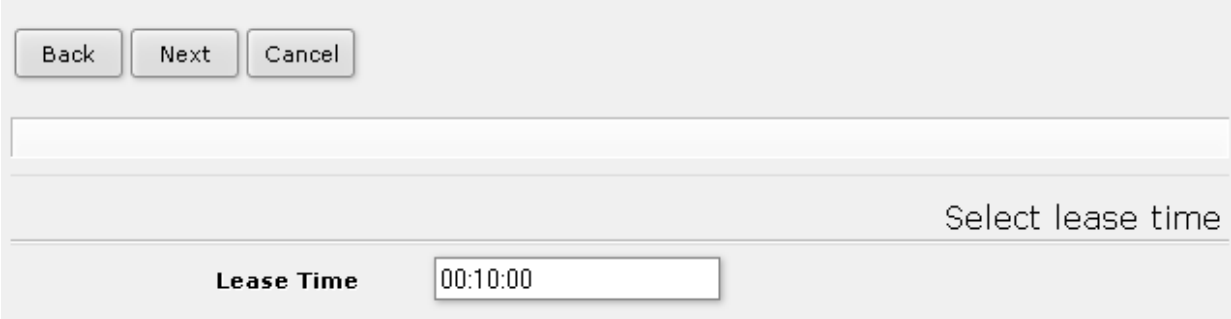
e. **DNS Servers : 192.168.252.132,** klik **Next**

(DNS ne'e sei hetan husi konfigurasaun ne'ebe mak antes ne konfigura ona, karik DNS lahanesan laiha buat ida, bele mos aumenta ho DNS ne'ebe mak baibain uja mak 8.8.8.8 no 8.8.4.4)



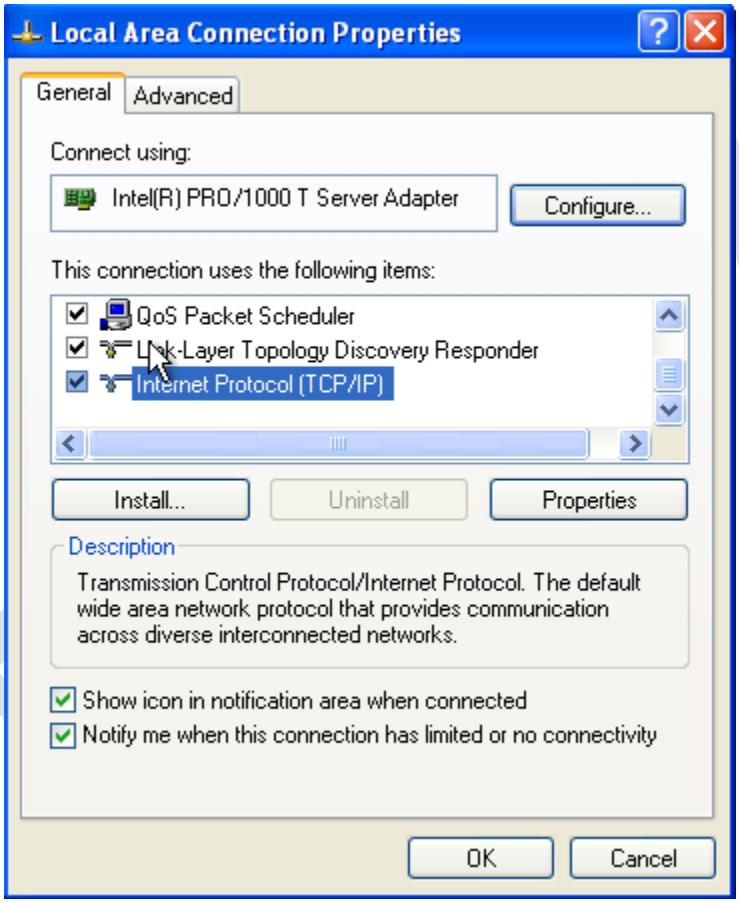
f. **Lease Time : 00.10.00,** klik **Next**

(husik default deit)



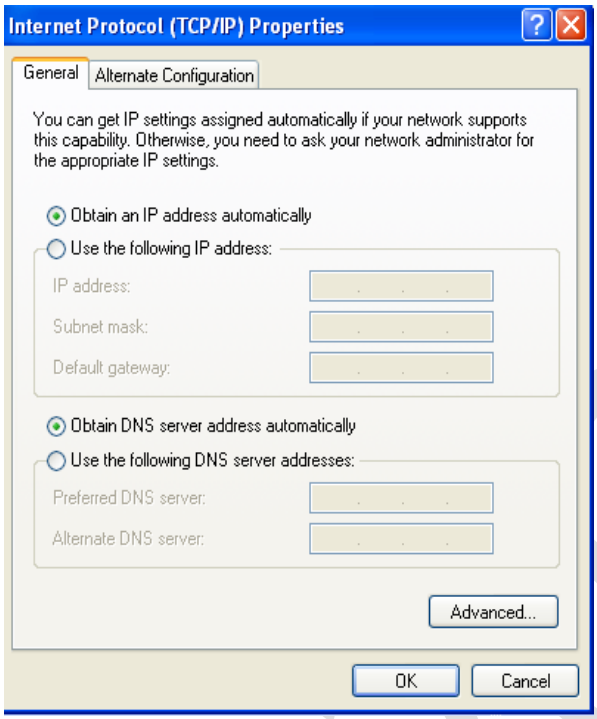
Rezultadu

1. **Windows XP**
   a. Klik **Start → Control Panel**
   b. Hili **Network and Internet Connections**
   c. Pois klik **Network Connections**
   d. Double Click ou Klik dala rua iha icon **Local Area Connection**
   e. Klik **Properties**, sei mosu informasaun husi **Local Area Connection**
   f. Scroll tun Hili **Internet Protocol(TCP/IP)**



g. Klik **Properties**, sei mosu tan janela foun hodi prense **IP Address**

h. Fo marka iha **Obtain an IP Addres Automatically**, pois klik **OK**

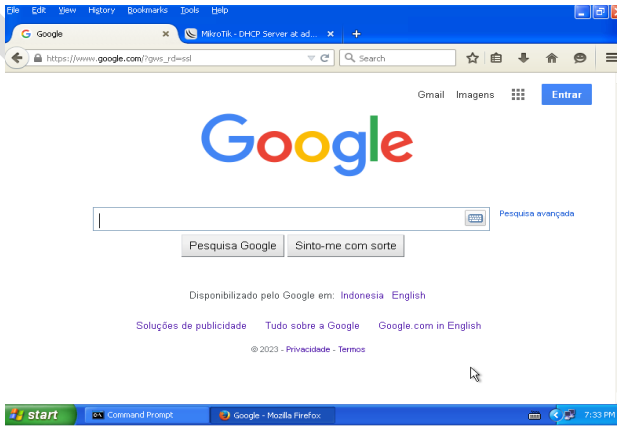


2. Koneksaun Rede Internet

Komputador client iha ne'e uja windows XP atu hatene katak komputdaor client windows XP hetan ona ou iha ona koneksaun ba rede internet ka seidauk?

a. Loke browser iha windows Xp koko asesu ba iha website ruma ezemplu www.google.com

b. Karik konfigurasaun los windows XP sei hetan rede internet no pagina husi google.com sei mosu



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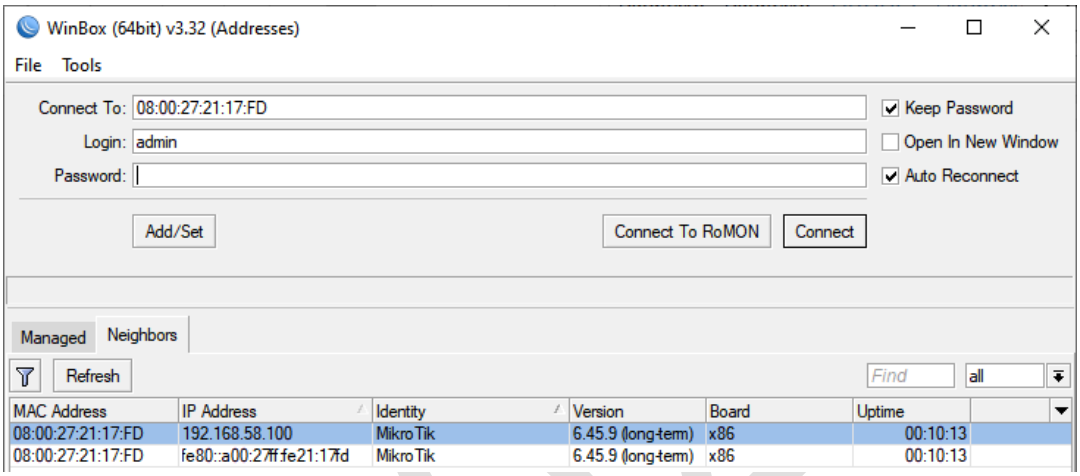
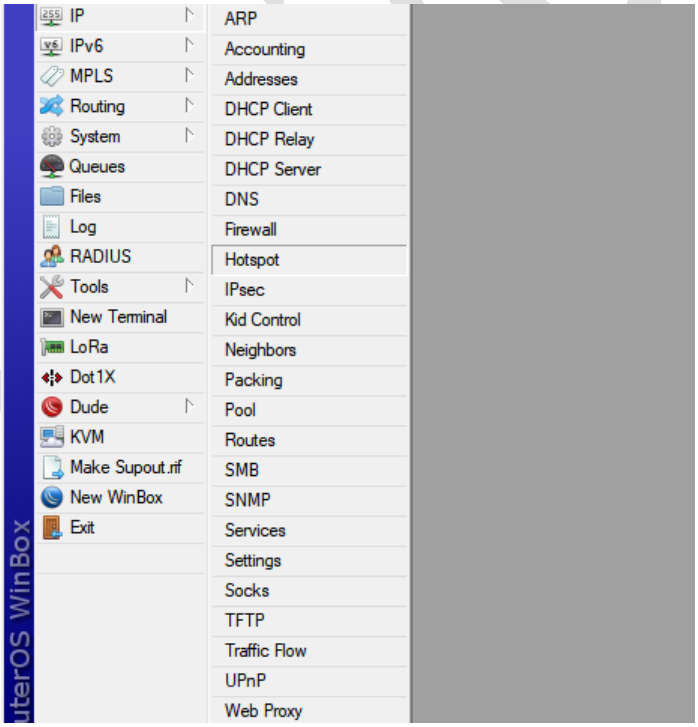
F. Konfigurasaun Hotspot Server Utiliza Aplikasaun Winbox

1. Iha Winbox hili **MAC Address** husi **Mikrotik**, klik **Connect**.

   Presija mos prense parte **Login** no **Password** karik iha mudansa, maibe tamba iha ne'e uza default login husi mikrotik, **Login : Admin** no **Password : husi mamuk**.



2. Iha **Mikrotik** klik iha Menu **IP → Hotspot**, sei mosu pajina husi **Hotspot Setup**



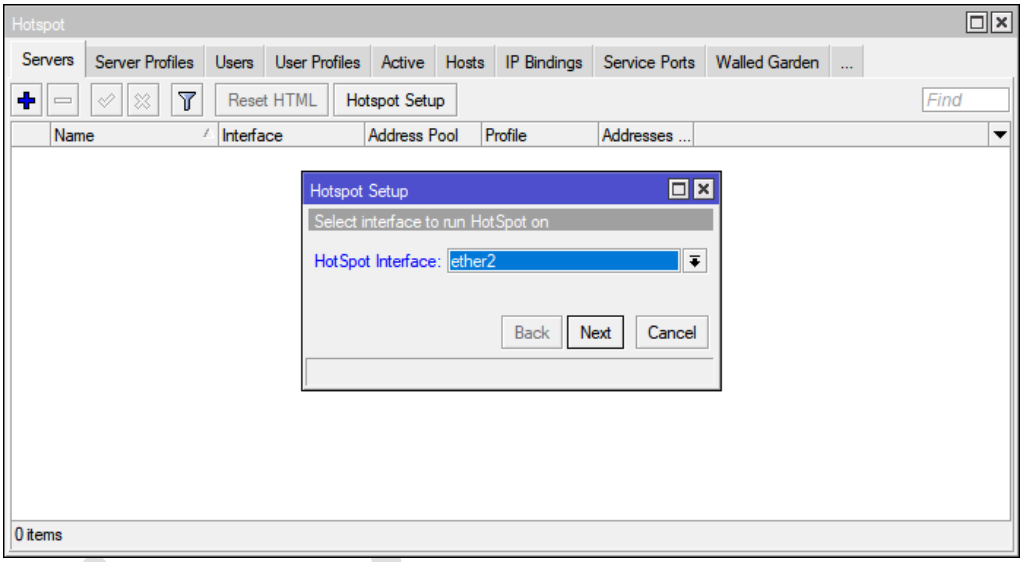
3. Iha pajina **Hotspot**, klik tab Menu **Servers** klik **Hotspot Setup**

   Konfigura tuir konfigurasaun tuir mai :

   a. Iha Hotspot Setup,

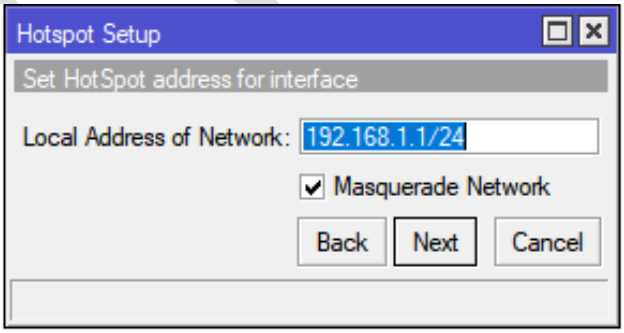
      Select interface to run Hotspot on

      HotsSpot Interface : ether2 (hili ether2, tamba hotspot sei konfiura iha parte interface ho naran ether2), klik **Next**.



   b. *Set Hotspot Address for Interface*

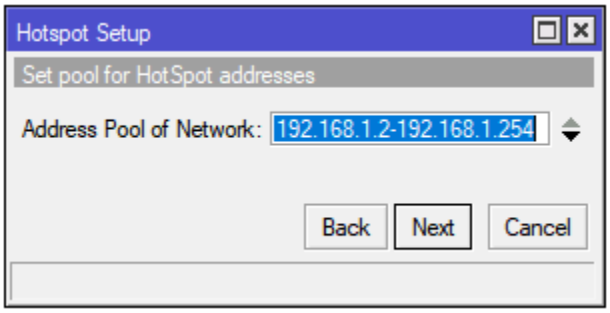
      Local Address of Network : 192.168.1.1/24 (ip address ho prefix /24 sei mosu automatikamente tamba ip address refere konfigura ona antes ne'e), klik **Next**.



   c. *Set Pool for Hotspot Addresses*

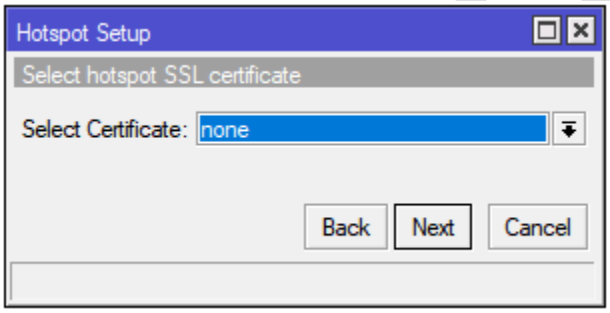
      Address Pool of Network : 192.168.1.2 – 192.168.1.254 (Mikrotik sei konfigura ou fo Pool bazeia ba IP Address ne'ebe mak konfigura antes ne'e, pool husi IP Address ho /24 maka 1-254, maibe iha ne'eba Pool Address

komesa husi 192.168.1.2 tamba IP Address 192.168.1.1 uza on aba iha Mikrotik iha Interface : ether2), klik **Next**



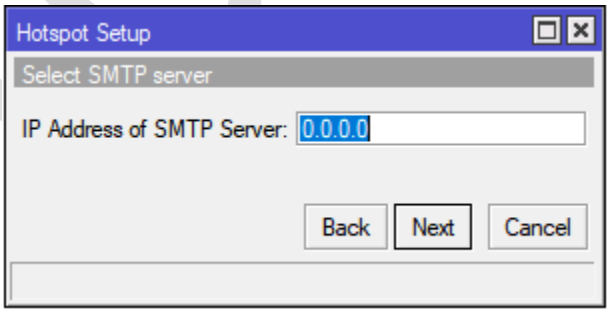
d. *Select Hotspot SSL Certificate*

Select Certificate : none (tamba laiha certificate hili none), klik **Next**



e. *Select SMTP Server*

IP Address of SMTP Server : 0.0.0.0 (wainhira iha rede refere iha konfigurasaun Email Server, bele hakerek IP Address husi Email Server, tamba laiha konfigurasaunn ruma konaba Email Server ita sei uza default konfigurasaun husi mikrotik), klik **Next**.

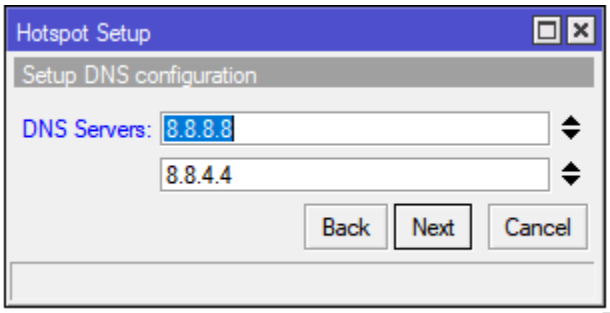


f. *Setup DNS Configuration*

DNS Server : 8.8.8.8

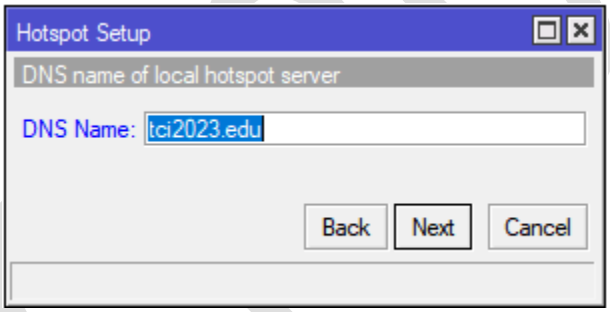
8.8.4.4

(aliende it abele uza DNS Server husi ISP, ita mos bele aumenta DNS Server husi google), klik **Next**



g. DNS Name of Local Hotspot Server

DNS Name : tci2023.edu (DNS Name refere sei mosu ou sei uja deit iha rede local husi hotspot ne'ebe mak ita kria), klik **Next**



h. Create Local Hotspot User

Name of Local Hotspot User : admin

Password for the User : admin

(Username refere username default wainhira ita konfigura hotspot mikrotik, ita-bot bele hakerek password tuir konfigurasaun idaidak nian, no Username refere mos sei uja wainhira login iha hotspot, ita mos bele kria tan Username seluk bazeia ba nesesidade), klik **Next**

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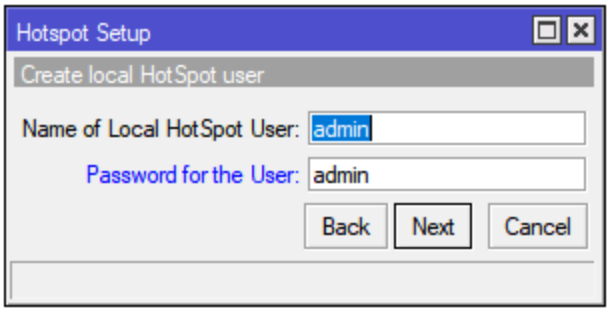
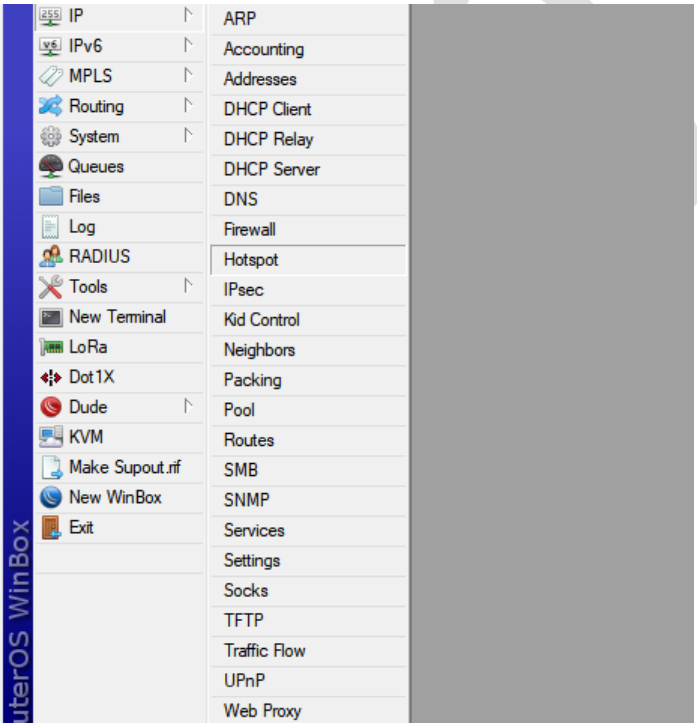


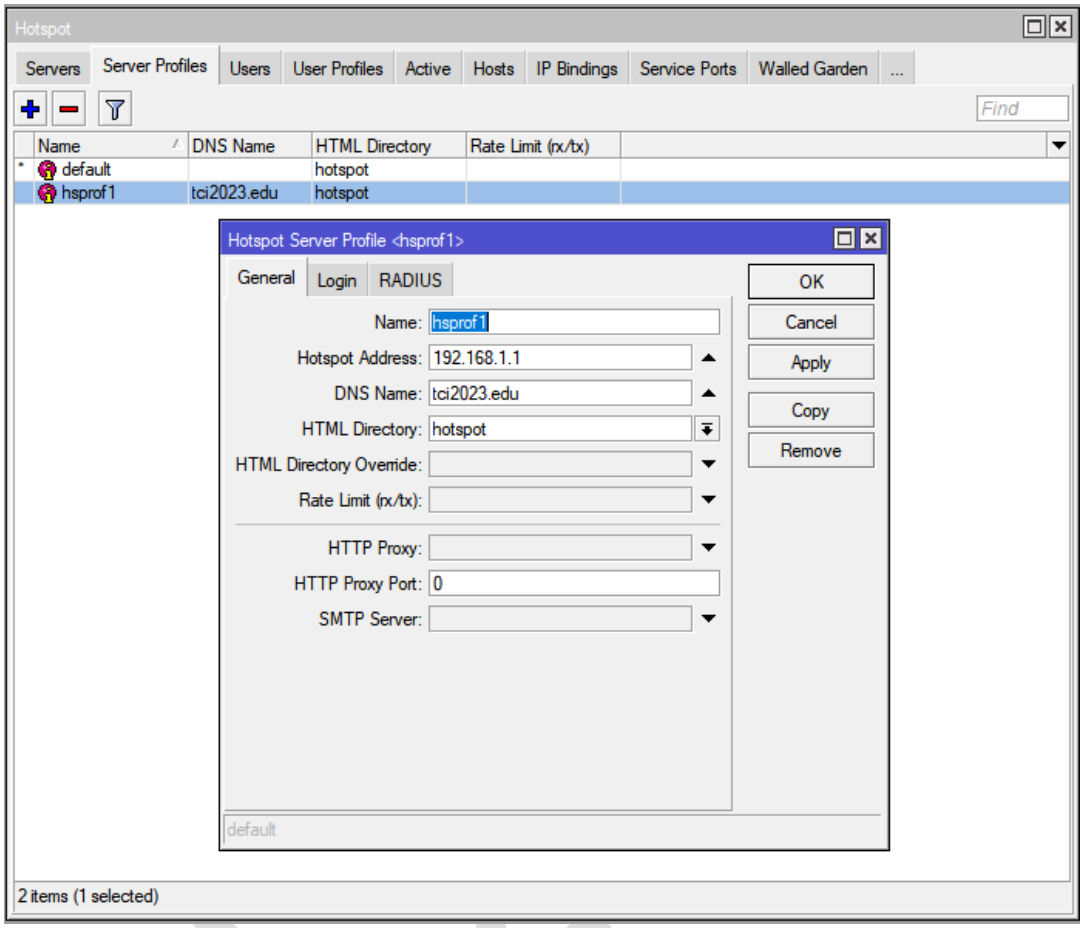
Setup has Complete Successfully, *signifika katak konfigurasaun Hotspot kompletu ho sussesu*

Konfigurasaun Hotspot

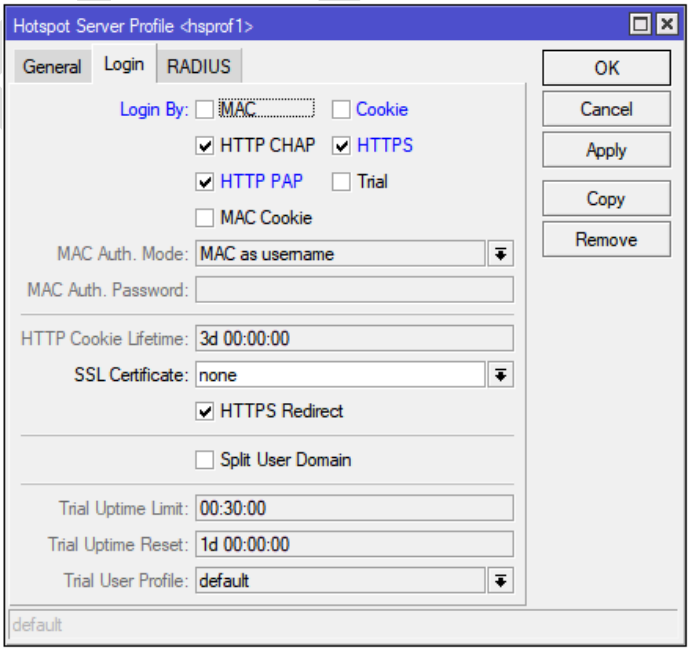
a. Iha **Mikrotik** klik iha Menu **IP → Hotspot**, sei mosu pajina husi **Hotspot**



b. Iha pajina Hotspot, klik tab Menu **Server Profiles**, double klik ou klik dalarua iha opsaun profile ho DNS Name **tci2023.edu** to'o mosu pajina Properties husi Hotspot Server Profile hanesan imagem tuir mai



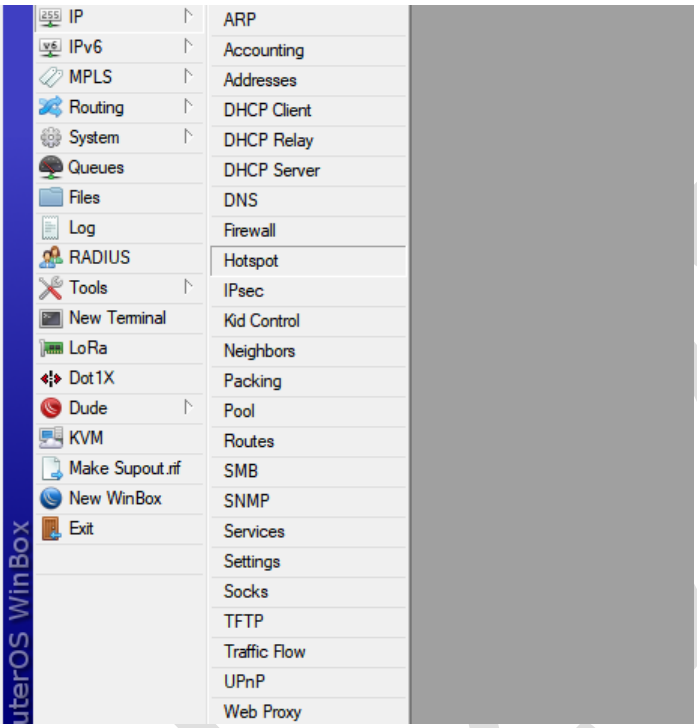
c. Klik iha tab Menu **Login**, ita sei halakon marka iha **Cookie** no fo marka iha **HTTP CHAP**, **HTTP PAP** no **HTTPS**, klik **Apply** pois **OK**



Rezultadu

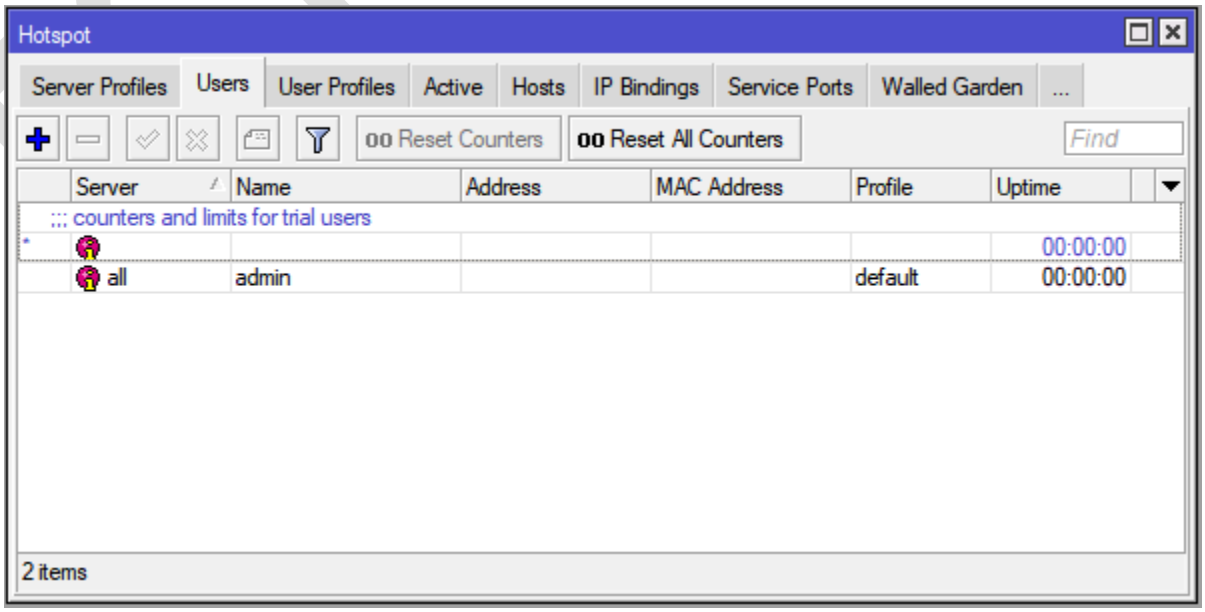
1. **Kria Username iha Hotspot**

   a. Iha **Mikrotik** klik iha Menu **IP → Hotspot**, sei mosu pajina husi **Hotspot**



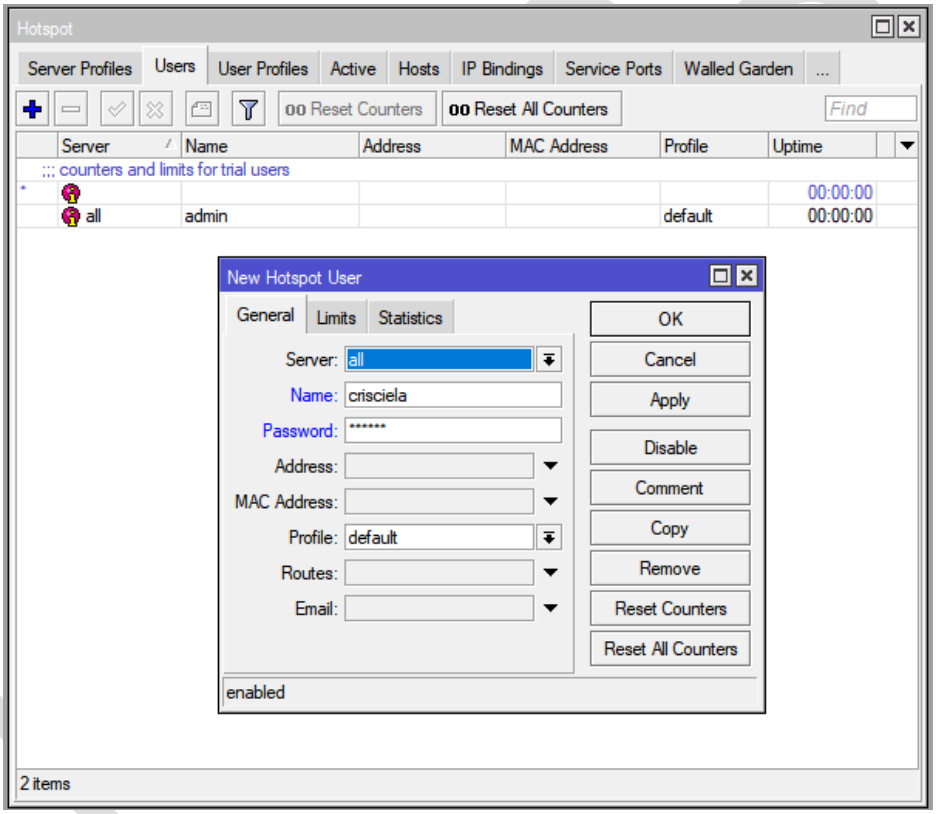
   b. Iha pajina Hotspot, klik tab Menu **Users**

   Sei iha username iha parte Name ne'ebe mak ita kria ho naran **admin**



c. Atu kria user foun iha hotspot hodi nune'e user refere ita abele fo ba client ne'ebe mak atu uja ou bele mos kria user foun ba ita nia an ho maneira hanesan tuir mai.

- Klik iha marka + iha sorin karuk leten
- Sei mosu pajina husi **New Hotspot User**
- Klik iha tab Menu **General**
- Iha parte **Name** prense user nia naran (ezemplu : **crisciela**)
- Iha parte **Password** prense user refere nia password (ezemplu : **123456**)
- Opsaun seluk husi default, klik **Apply** pois **OK**



d. Karik atu kria tan user seluk, halao konfigurasaun ne'ebe hanesan to remata.

2. Koneksaun Rede Internet

a. Loke browser iha Windows Xp koko asesu ba iha website ruma ezemplu www.google.com

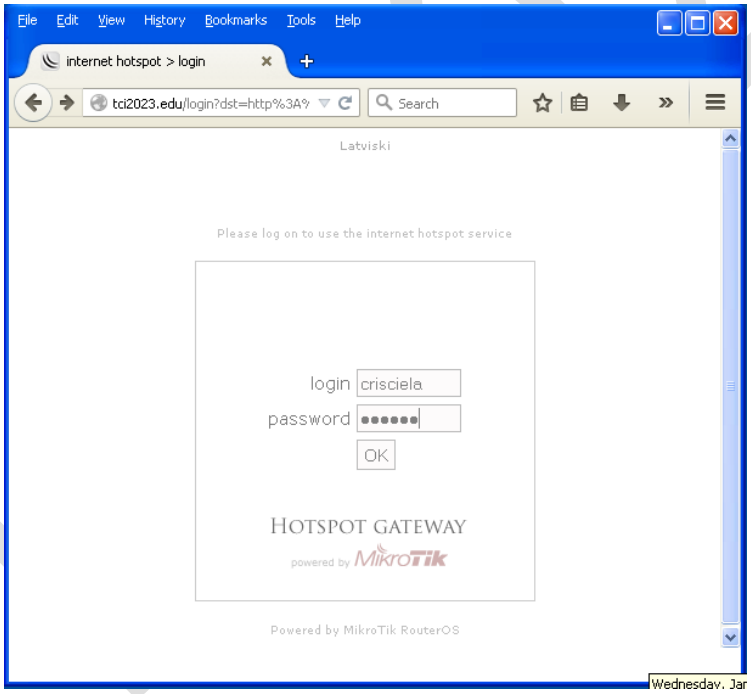
b. Browser sei lori ita ba iha pagina Login husi hotspot

(Router Mikrotik liu husi konfigurasaun Hotspot sei halo konfirmasaun ba iha client ne'ebe mak atu asesu linha internet, konfirmasaun refere sei hare liu ba autentikasaun husi client hanesan username ho password. Client ne'ebe iha username ho password ou client ne'ebe mak iha autorizasaun husi Router mak bele asesu ba linha internet)

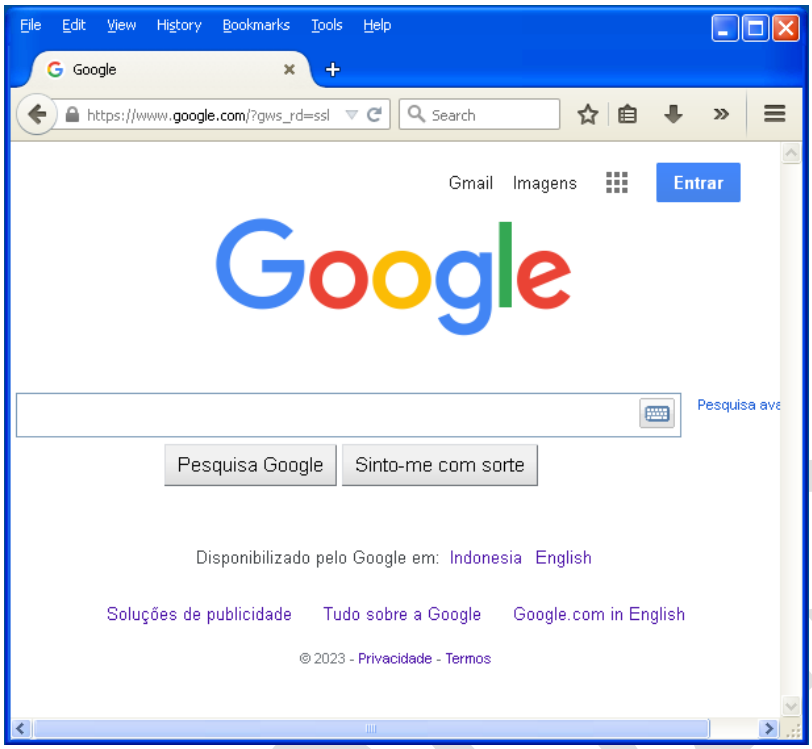
c. Prense username ho password ne'ebe mak ita kria ona

Login : crisciela

Password : 123456



d. Karik prosesu autentikasaun los, browser sei lori ita ba pajina ne'ebe mak ita husu (request) mak www.google.com



e. Bele mos hare informasaun konaba username ne'ebe ita uja Login ho maneira hakerek **tci2023.edu** iha TaskBar husi browser.

OBRIGADO